Audit qualifications for the accompanying research draft

A-SNN Evidence-1. Updated 14 October 2026.

Scope of this revision. This front matter accompanies the original ten-page draft. It adds the limitations identified by the project audit. Every original manuscript page, graph, table and numerical result follows unchanged. The work remains a presentation draft, not a peer-reviewed publication.

Historical benchmark-test exposure. SHD and SSC official test results informed earlier exploratory architecture choices. Validation-selected checkpoints protect epoch selection in the present runner, but do not undo that project-level test exposure. The reported confidence intervals summarize training-seed variation on these known benchmarks, not a previously untouched external holdout.

State-intervention interpretation. The selected hybrid depends on its original sample-state pairing. Global state derangement usually changes class as well as sample. It does not isolate identity within a class. Zero-state deletion jointly removes spike traces and analog final membrane, so it does not isolate spikes from analog state or establish timing-specific coding. Separately retrained and analog-state controls require a new registered study.

Results and provenance remain bounded. Both original runs report causal_audit_only. Accuracy and compiler numerical-fidelity gates fail. The repeated ten seeds do not form twenty independent observations or certify independent clean-clone replication. The current hardened v2 code is a new implementation version, not a retroactive replacement of the source that generated the original records.

Research readiness. Human-reviewed decision lock, source-complete public release, independent replication and the cause of the two finite compiler discrepancies remain pending. No biological, energy-efficiency or competitive-superiority claim follows from this draft.

Authority: gen5/docs/PROJECT_AUDIT_2026_10_14.md and EVIDENCE1_HARDENING_2026_10_14.md. Original experimental archives and registered margins remain unchanged.

State Dependence Without Competitive Superiority: A Paired Audit of Residual LIF Models on Spiking Audio

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Abstract

A temporal state can be important to a trained classifier without making that classifier competitive with conventional alternatives. We examine this distinction in the A-SNN project using a residual leaky integrate-and-fire (LIF) model on Spiking Heidelberg Digits (SHD) and Spiking Speech Commands (SSC). An internally frozen protocol compares four parameter-budget-matched architectures across ten paired seeds per dataset, with validation-selected checkpoints and fixed-checkpoint state interventions. In the primary run, residual-LIF accuracy is 84.06% on SHD and 56.48% on SSC. Removing state reduces accuracy by 9.43 and 13.64 percentage points, respectively; replacing state with globally deranged sample partners reduces it by 4.04 and 8.87 points. All four paired 95% confidence intervals exclude zero. However, the registered one-point noninferiority condition fails: for example, residual LIF trails the SSC temporal convolutional network by 1.96 points (95% interval [-2.82, -1.10]). Compilation yields approximately tenfold median inference speedups, but two residual checkpoints exceed the registered numerical tolerance despite identical validation class predictions. A second execution of the same ten seeds reproduces the aggregate outcome but is not certified as independent external replication. These results support a model-internal, sample-specific state-dependence finding and a reusable audit workflow, not competitive superiority, energy efficiency or biological fidelity.

Keywords: spiking neural networks; residual state; auditory classification; paired experiments; state interventions; numerical fidelity; reproducibility.

1. Introduction

Spiking neural networks (SNNs) use discrete events and evolving internal states to process information. Surrogate-gradient methods make such models trainable using gradient-based optimization despite hard thresholds [2]. Auditory spike datasets provide a practical setting for evaluating these methods [1]. Nevertheless, a successful SNN experiment leaves several distinct questions unanswered: does its state affect the prediction, is that effect specific to the input sample, does the whole model match strong dense alternatives, and does its execution backend preserve the intended computation?

This paper separates those questions. A residual model combines direct temporal-convolution features with a spiking state branch. We intervene on that branch at a fixed selected checkpoint, then independently compare the complete architecture with matched dense controls. We also evaluate compiled inference against the identical eager checkpoint. These comparisons address different claims and must not be collapsed into a single “SNN advantage.”

The A-SNN project previously explored multiple adaptive mechanisms. The present study does not attempt to validate that entire research program. It consolidates a narrow temporal-audio finding under the

Evidence-1 protocol. The study's practical contribution is a transparent separation of positive state-intervention evidence from unsuccessful accuracy and systems gates. We do not claim to introduce the first residual SNN, a new neuron model, or a new causal-inference method.

Our contributions are: (i) a ten-seed, validation-selected comparison on two official auditory spike benchmarks; (ii) globally deranged, fixed-checkpoint state controls that preserve the sample's direct feature path; and (iii) a numerical-and-throughput audit that reports failed conditions rather than promoting class agreement alone as compiler equivalence.

2. Background and scope

Cramer et al. introduce the Heidelberg spiking datasets and an audio-to-spike conversion procedure intended to support systematic SNN evaluation [1]. SHD distinguishes spoken digits in English and German, while SSC is based on speech commands. We use the official spike files and a fixed coarse temporal encoding; published accuracies obtained with different preprocessing are not treated as matched baselines.

Surrogate gradients replace the derivative of the hard threshold during backpropagation without replacing its forward event operation [2]. Dense temporal convolutions and gated recurrent units provide relevant conventional controls [3, 4]. Our TCN is a compact two-layer residual dilated-convolution implementation, not a complete reproduction of the architecture or experiments of Bai et al. The GRU is a single-layer last-hidden-state classifier. AdamW supplies the common optimizer [5].

The word causal is used here in a restricted computational sense: interventions change selected model features while holding the trained checkpoint and the direct path fixed. This does not identify a biological mechanism, a population-level treatment effect, or an advantage over every possible retrained architecture. In particular, zero-state intervention is not equivalent to training a new direct-only model from scratch.

3. Methods

3.1 Datasets and encoding

Each sample is represented as X□{0,1}^64×700. An event at time t is mapped to bin b=□ 64t/D□, where D=1.4 seconds for SHD and D=1.0 second for SSC. Valid events set the corresponding channel/bin entry to one. Multiple events in the same bin/channel collapse to binary occupancy; this is not a count-preserving representation. Events outside the configured time/channel range are excluded by the encoder.

SHD uses a stratified 20% holdout from the official training set with split seed 2026. SSC uses its official validation split. We use complete encoded splits, without sample limits. Table 1 gives the actual split sizes recorded in both bundles. Final test data are not used for checkpoint selection. The temporal convolutions use symmetric padding; the model is evaluated as an offline whole-utterance classifier, not a streaming-causal predictor.

Table 1. Encoded benchmark splits from retained provenance. Both datasets use 700 channels and 64 binary temporal bins.

Dataset	Classes	Duration (s)	Train	Validation	Test
SHD	20	1.4	6,526	1,630	2,264
SSC	35	1.0	75,466	9,981	20,382

3.2 Residual LIF architecture

A width-five temporal convolution produces currents I_t=Conv1D(X)_t and the direct trace d_t=ReLU(I_t). For each channel, the state branch follows

$$\widetilde{u}_t = \lambda \odot u_{t-1} + I_t, \quad u_0 = 0$$

$$s_t = H(\widetilde{u}_t - \theta)$$

$$u_t = \widetilde{u}_t - \theta \odot s_t$$

Here H(v)=1[v≥0], λ=σ(l) is a learned leak, and θ=max(softplus(q),10^-3) is a learned threshold. Initial leak and threshold are 0.90 and 1.0. The hard-threshold surrogate derivative is (1+10|v|)^-2. The recurrence integrates convolutional currents; there is no learned recurrent neuron-to-neuron weight matrix in this branch.

Let P concatenate temporal mean-pooling windows at levels {1,2,4,8}, giving 15 windows per trace. The linear classifier consumes

$$z = [\mathcal{P}(d); \mathcal{P}(s); u_{64}/\theta], \quad \hat{y} = \operatorname{argmax}(Wz + b)$$

For branch width C, the readout has 31C inputs. The model is a hybrid of direct analog features, discrete spikes and an analog final membrane state; it is not an all-spike-output pipeline.

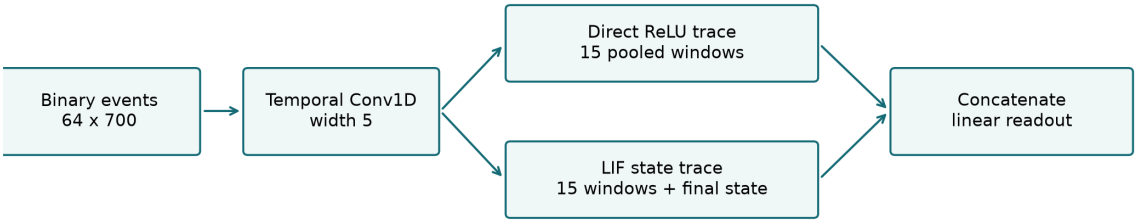


Figure 1. Representation boundary used by the interventions. Zeroing or exchanging state leaves the original sample's direct trace and classifier weights unchanged.

3.3 Dense controls and training

The controls are a one-layer Conv1D with the same pooling levels, a two-layer residual TCN with width-five input convolution and width-three dilation-two hidden convolution, and a single-layer GRU with a linear last-hidden-state readout. Widths are chosen to stay at or below 133,631 trainable parameters and at or above 95% of that ceiling. Exact counts appear in Table 2; all parameters are active dense parameters. Matching parameter budgets does not match FLOPs, wall-clock training time, receptive field or optimization difficulty.

Each architecture uses seeds 310–319, 15 fixed epochs, batch size 256, cross-entropy loss and AdamW with learning rate 0.003 and weight decay 0.0001. There is no per-arm tuning, scheduler or early stopping. The earliest checkpoint attaining the highest validation accuracy is selected. Epoch minibatch permutations are paired across architectures. The seed identifiers are shared, but different architectures do not have identical initialization tensors. Training is eager; compilation is an inference measurement after checkpoint selection.

3.4 Fixed-checkpoint interventions

For each selected residual checkpoint we evaluate:

- Full: the sample's direct and state features.

- Zero state: zero pooled spike-state and final membrane features, while retaining the original direct trace and the full model's readout.

- Shuffled state: exchange the spike trace and final membrane with those generated by a different sample, retaining the original direct trace and label.

The partner mapping is a dataset-wide derangement: it contains no fixed points and may cross minibatches, including the tail batch. Three mappings use partner seeds s+80000+r, with r□{0,1,2}. Their accuracies are averaged within each training seed before computing the paired difference. The three mappings are not treated as three independent training runs. No intervention is retrained, and shuffle partners are not conditioned on class labels.

3.5 Uncertainty and internally frozen gates

For seed-level differences Δ_s, we report

$$\overline{\Delta} \pm t_{0.975,9}\, s_{\Delta}/\sqrt{10}$$

Accuracy summaries use the analogous seed-level Student-t interval. Differences are reported in absolute percentage points (pp). The seed is the experimental unit. These intervals describe variation across the selected training seeds on fixed data splits, not uncertainty over a newly sampled population of speakers. Intervals are marginal, not simultaneous familywise intervals.

The dated internal protocol separates four requirements:

- State contribution: both full-minus-zero and full-minus-average-shuffle lower bounds exceed zero on both datasets.

- Accuracy noninferiority: the residual-minus-baseline lower bound exceeds -1 pp against each dense control on both datasets. This is not a two-sided equivalence or superiority test.

- Systems: every residual checkpoint has identical compiled and eager validation class predictions, maximum absolute logit error ≤10^-4, and compiled/eager throughput ratio greater than one.

- Replication: independent clean-clone provenance and a human-reviewed aggregate decision lock.

The specification was internally frozen before the supplied confirmation runs according to the dated project record. It was not externally registered in a public timestamped registry. A pass in one gate cannot compensate for a failure in another.

3.6 Execution and artifact audit

Both retained executions report NVIDIA L4 hardware, Python 3.13.15, PyTorch 2.11.0+cu130 and CUDA 13.0. Compiler probes use torch.compile with fullgraph=True and mode="reduce-overhead". Fidelity is checked on the complete validation split, including tail shapes. Throughput is measured on the first batch of 256 validation examples, with five warmups and the median of three 20-iteration timing repeats, using device synchronization. Transfer and data-loading costs are outside this inference measurement. It is not an end-to-end serving benchmark or a multi-GPU measurement.

We verify the supplied outer SHA-256 manifests, unique run keys, per-run record equality, recorded source/data/protocol agreement, and all reported seed-level means, intervals and gates. The audit parses JSON and hashes archive bytes; it does not execute embedded code or load checkpoint pickles. Recorded source fingerprints are preserved in the supplement. Numerical consistency alone cannot certify who ran a clean clone.

4. Results

4.1 Matched-architecture accuracy

Table 2 and Figure 2 show the primary run's accuracy and the repeat execution's means. Residual LIF reaches 84.06% on SHD and 56.48% on SSC. The SSC TCN reaches 58.44%, while the GRU reaches 57.44%. On SHD, the GRU reaches only 40.41%; that weak result is a limitation of this particular common-budget training recipe, not evidence that GRUs are generally unsuitable for auditory classification.

Table 2. Test accuracy in percent. SD and 95% intervals are across ten seeds of the primary run; the last column is the same-seed repeat mean. No seeds from the executions are pooled.

Dataset	Model	Parameters	Mean +/- SD (%)	95% interval (%)	Repeat mean (%)
SHD	Residual LIF	131,956	84.06 +/- 1.15	[83.24, 84.89]	84.06
SHD	Conv1D	133,055	83.24 +/- 2.01	[81.80, 84.68]	83.24
SHD	TCN	132,756	83.13 +/- 1.43	[82.11, 84.15]	83.22
SHD	GRU	133,420	40.41 +/- 3.24	[38.09, 42.73]	40.41
SSC	Residual LIF	133,087	56.48 +/- 0.66	[56.01, 56.95]	56.48
SSC	Conv1D	132,893	49.77 +/- 2.27	[48.14, 51.39]	49.77
SSC	TCN	131,971	58.44 +/- 0.98	[57.74, 59.15]	58.44
SSC	GRU	131,819	57.44 +/- 0.66	[56.97, 57.91]	57.44

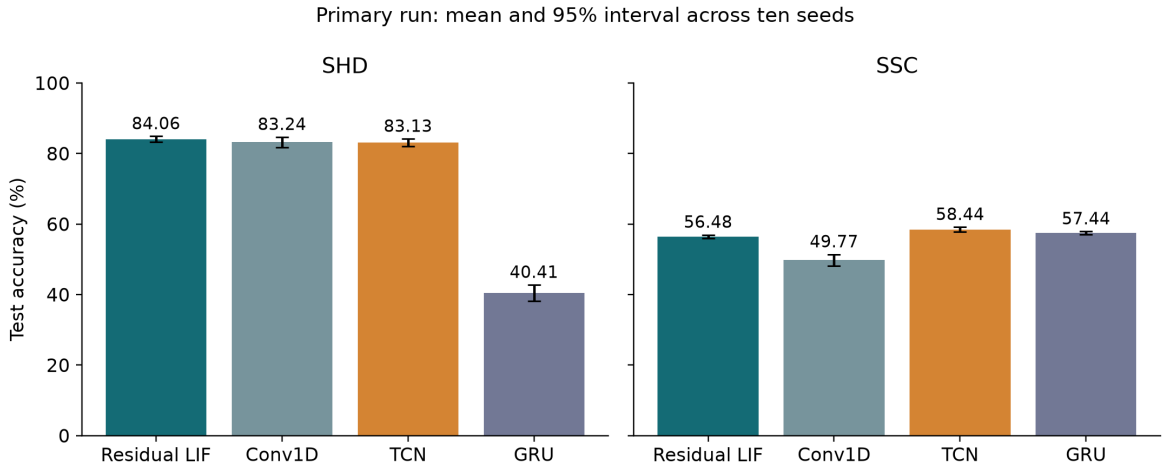


Figure 2. Primary-run means with 95% seed-level intervals. The full zero-to-100 scale avoids overstating small accuracy differences. Overlap of marginal intervals is not the registered paired noninferiority test.

4.2 State contribution and sample specificity

All four state-contribution intervals have positive lower bounds (Table 3, Figure 3). On SHD, the mean full-minus-zero effect is 9.43 pp (95% interval [7.36,11.51] pp), and the full-minus-shuffled effect is 4.04 pp ([3.19,4.89] pp). On SSC, the corresponding effects are 13.64 pp ([10.91,16.37] pp) and 8.87 pp ([8.27,9.46] pp).

The state branch therefore supplies information used by these trained readouts, and correct sample-to-state correspondence matters. However, exchanging state also breaks learned feature compatibility. The effect cannot isolate spike timing from rate, final membrane magnitude or co-adaptation with the direct branch. It does not show that an appropriately retrained direct-only architecture would suffer the same loss.

Table 3. Fixed-checkpoint state effects, in percentage points. Shuffle accuracy is averaged over three global derangements within each of ten seeds. Both runs produce the same reported effects.

Dataset	Contrast	Mean (pp)	Paired 95% interval (pp)	Gate
SHD	Full minus zero state	9.43	[7.36, 11.51]	Pass
SHD	Full minus shuffled state	4.04	[3.19, 4.89]	Pass
SSC	Full minus zero state	13.64	[10.91, 16.37]	Pass
SSC	Full minus shuffled state	8.87	[8.27, 9.46]	Pass

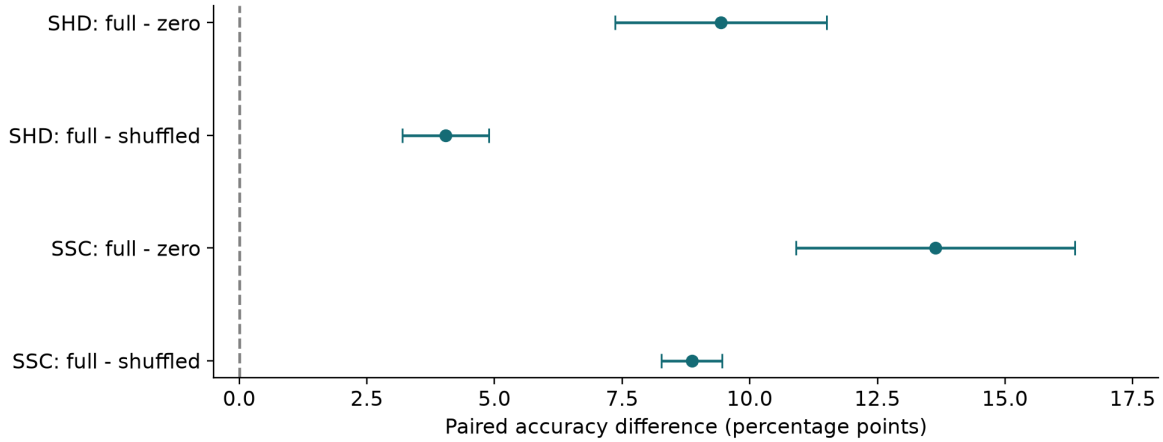


Figure 3. Mean paired effects with 95% Student-t intervals. Every interval lies above the zero-effect boundary; the finding is restricted to these fixed-checkpoint interventions.

4.3 Accuracy gate: a negative result

The complete accuracy gate fails on both datasets (Table 4). On SHD, residual-minus-Conv1D has a positive mean of 0.82 pp, but its lower bound is -1.13 pp, below the registered -1 pp margin. This failure to establish noninferiority does not prove that residual LIF is inferior to Conv1D.

On SSC, residual-minus-TCN is -1.96 pp with interval [-2.82,-1.10] pp, entirely below the margin in the primary run. Residual-minus-GRU also fails the lower-bound condition. Figure 4 shows all three SSC comparisons. The repeat SSC TCN difference is -1.96 pp, with interval [-2.80,-1.13] pp; the decision is unchanged. Passing the causal controls therefore does not establish dense-baseline parity under the registered recipe.

Table 4. All registered paired dense-baseline comparisons in the primary run. Passing requires the lower bound to be strictly greater than -1 percentage point.

Dataset	Contrast	Mean (pp)	Paired 95% interval (pp)	Gate
SHD	Residual minus Conv1D	0.82	[-1.13, 2.78]	Fail
SHD	Residual minus TCN	0.93	[-0.40, 2.26]	Pass
SHD	Residual minus GRU	43.65	[40.71, 46.60]	Pass
SSC	Residual minus Conv1D	6.71	[5.01, 8.41]	Pass
SSC	Residual minus TCN	-1.96	[-2.82, -1.10]	Fail
SSC	Residual minus GRU	-0.96	[-1.66, -0.26]	Fail

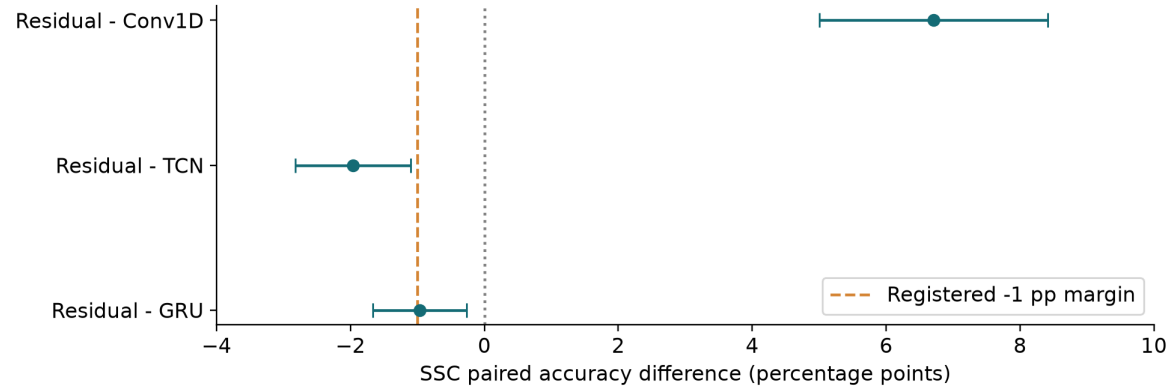


Figure 4. SSC accuracy contrasts. The dashed line is the registered -1 pp margin, not zero. TCN and GRU fail the required lower-bound condition. Table 4 includes the SHD comparisons as well.

4.4 Compiler speed and fidelity

Residual checkpoints show median speedups of 10.55× on SHD and 10.74× on SSC in the primary run, and 10.75×/11.06× in the repeat (Table 5). These are observed steady-state, same-model inference ratios at batch 256, not a speed advantage over the TCN or GRU and not an energy measurement.

All 20 residual checkpoints in each execution have identical eager and compiled validation class predictions. Nevertheless, SHD seed 310 has maximum absolute logit error 3.1518936×10^-4 and SSC seed 311 has 9.7311020×10^-2, exceeding 10^-4. The same two exceptions appear in both executions. The remaining 18 residual checkpoints pass the tolerance. Consequently, the registered systems gate fails despite the speedups (Figure 5). The root cause of these discrepancies has not been established. We do not infer it from the speed or accuracy records.

GRU compilation is recorded as unsupported in the supplied execution; its eager training and accuracy evaluation are complete. This compilation failure does not invalidate its role as an accuracy control. A different backend would require a separate, documented systems evaluation.

Table 5. Residual checkpoint inference only. Each row summarizes ten seeds. All class predictions agree; numerical tolerance failures keep the systems gate closed.

Dataset	Execution	Median speedup	Logit tolerance passes
SHD	Primary	10.55x	9/10
SSC	Primary	10.74x	9/10
SHD	Repeat	10.75x	9/10
SSC	Repeat	11.06x	9/10

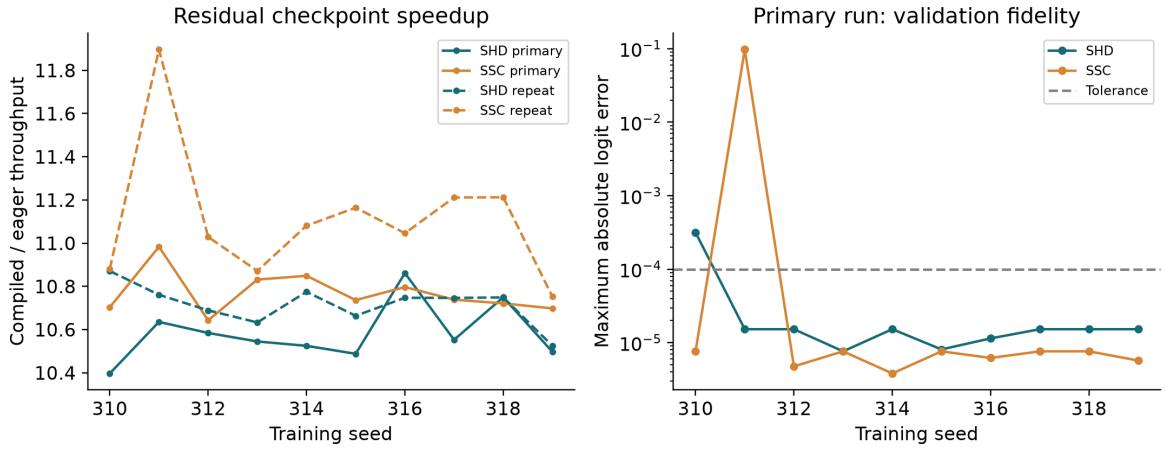


Figure 5. Speedups from both executions and numerical errors from the primary run. The errors use a logarithmic scale. Matching class labels does not satisfy the stricter logit-fidelity contract.

4.5 Repeat execution and aggregate decision

Both executions contain all 80 architecture/dataset/seed fits and the same protocol signature. Their recorded source fingerprints, raw data and encoded cache provenance agree. Residual, Conv1D and GRU mean accuracies agree; TCN means differ slightly (SHD 83.13% versus 83.22%, SSC 58.4408% versus 58.4393%). We do not claim bitwise reproducibility of all training. Both executions return causal_audit_only, with state contribution passing and accuracy/systems gates failing on both datasets.

The repeats reuse the same ten seeds and fixed splits. They do not create twenty independent seed observations or establish a second independent research team's reproduction. Both machine summaries explicitly leave the clean-clone replication gate pending, and the formal human-reviewed decision lock has not been recorded. This paper is therefore a presentation draft of the retained evidence, not a completed claim of independent replication.

Recorded training-plus-validation time summed across 80 fits is 66.70 minutes in the primary execution and 67.52 minutes in the repeat. Those fields exclude data preparation, downloads, post-training interventions, compilation and other notebook overhead; they are not total job durations.

5. Discussion and limitations

Positive dependence, negative competitiveness. The results support a useful distinction between information used by a trained model and overall model quality. The residual state is not inert: both removal and sample reassignment reduce performance consistently. Yet the complete model fails the predeclared dense-baseline condition. This makes the audit itself, rather than a superiority claim, the defensible contribution of the current study.

Restricted intervention interpretation. State manipulation changes the readout's feature distribution. The full-minus-shuffled effect shows sensitivity to state/sample correspondence, but does not isolate temporal coding or eliminate feature-co-adaptation explanations. Future rate-preserving perturbations, analog-state controls and separately retrained ablations would address different questions and require new frozen protocols.

Baseline optimization and input information. One common recipe is transparent but cannot establish that each architecture is near its optimum. The especially weak SHD GRU result merits a validation-only diagnostic. Binary occupancy at 64 bins discards event counts and finer timing. The present results are not direct comparisons with published models using different channel compression, binning, augmentation, readouts or training/search budgets.

Numerics and execution. The speedups compare compiled and eager implementations of the same residual model after warmup. They do not establish lower training cost, end-to-end latency, hardware-sparse execution or low-energy operation. All model parameters remain dense, and the compiler-fidelity tolerance fails at two checkpoints. Further numerical investigation is needed before describing this compiled path as validated under the registered contract.

Statistical and provenance boundaries. Only ten seed observations, two fixed datasets and one documented GPU type are studied. Marginal seed-level intervals are not speaker-population confidence intervals or familywise confidence statements. Manifest and summary checks establish internal consistency, not external execution independence. The public Git base recorded by the notebook predates the embedded restart implementation; a source-hashed clean-clone release is still needed for a complete reproducibility artifact.

6. Conclusion and next experiments

Under the internally frozen SHD/SSC protocol, residual LIF state contributes to fixed-checkpoint predictions and its sample-specific correspondence matters. The same model does not satisfy the registered accuracy-parity condition against all dense controls. Compilation improves observed steady-state inference throughput but does not satisfy numerical fidelity at every checkpoint. A repeat of the same seeds preserves this aggregate outcome without establishing independent external replication.

The immediate work is to finalize the evidence review and decision lock, publish a source-complete audit artifact, and diagnose compiler fidelity on frozen validation checkpoints. Only after that decision should new timing/readout comparisons be preregistered. A matched FiTS reference, explicit-context routing and small connectome-prior experiments are future work, not results of this paper. No change to the current protocol should be selected using these already inspected test outcomes.

Authorship, support and presentation status

The author names, student identifiers and departmental affiliations are provided by the presenters. Author order, institutional approval, contribution statements and acknowledgments should be confirmed with the research mentor before external submission. No grant, accepted publication, energy measurement or institutional endorsement is asserted in this draft. The manuscript reports existing artifacts; no additional training was performed to prepare it.

References

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Evidence provenance and reproducibility checklist

This appendix is an audit supplement, not an additional experiment. It records the supplied bundles used to generate all numerical tables and plots. The first execution is used for primary uncertainty reporting; the second is a same-seed repeat. The generation script recomputes the statistics directly from per-seed JSON records and checks them against the stored summaries to 10^-10 absolute tolerance.

Primary bundle: research_restart_results.zip.

Primary SHA-256: 9c97119530f9396dcdc889e5959c9e72528f48915a3e2e281e05c2c087e9b702

Repeat bundle: research_restart_replication_01.zip.

Repeat SHA-256: f254b552374cb20c695ec5173438e166e2d9a954c98399703869ed9a98b58f41

Shared protocol signature: 5684278f73ff052064cc6be14d2988a94331a74ca01be99662e46ff0c5ed40c3

Recorded public Git base: ee150bd1b0051830f8e3e959805374f4efac9deb

The accompanying evidence_audit.json retains the protocol, per-seed records, source fingerprints, data provenance and environment. verified_metrics.json provides a compact numerical summary; the four PNG charts are supplied separately for presentation use. The standalone LaTeX source contains the numerical plot coordinates, so the PDF does not depend on external graphics files.

The canonical local study sources are gen5/docs/EVIDENCE1_AUDIO_PREREGISTRATION.md, gen5/docs/PRIMARY_EVIDENCE_TRACK_ROADMAP.md, and gen5/ammc_gen5/evidence1_audio.py, including its loader, model and state-extraction dependencies. Graphify is only a derived retrieval layer. The pre-run status text in project documents is not substituted for the supplied result records.

Before treating this draft as a submission-ready research artifact:

- Confirm authorship, institutional wording and the mentor's review.

- Check the presentation venue's format, originality and AI-assistance policies.

- Retain the unchanged two source ZIPs and their checksums.

- Publish the exact implementation with its source fingerprints and environment instructions; the old Git base alone is insufficient.

- Complete independent clean-clone provenance review and document the human-reviewed decision without converting a failed gate into a pass.

- Keep any later compiler repair, hyperparameter change or architecture experiment separate from the frozen Evidence-1 results.